Display unit failures and reconfiguration

flyPad failures for the A32NX ND and ECAM display units and the eight A380X CDS display units: the failed display goes blank, with the A380X CDS ... DU FAULT alert; both aircraft move a lost display to another display unit (A32NX: PFD/ND XFR pbs, ECAM/ND XFR selector, held system page keys; A380X: PFD/ND and DU RECONF pushbuttons, the EWD and the SD also on the ND display unit); the A380X hosted EWD and PFD now hidden when not shown, and the CAPT PFD on the CAPT FMS side

Aircraft	A32NX and A380X
Where to find it	flyPad > Failures > ATA 31 Indicating / Recording System; the display units; A32NX PFD/ND XFR pbs, brightness knobs, SWITCHING panel ECAM/ND XFR, ECAM control panel; A380X E/WD and SD STATUS; A380X PFD/ND and DU RECONF pushbuttons
Branch	`core/display-unit-failures`, based on `core/nd-weather-terrain` (stacked: needs core/nd-weather-terrain merged first, the radar / terrain gauge goes blank with a failed ND). The A32NX reconfiguration also needs the cockpit model companion in aircraft-large-files (fork branch `feat/a32nx-du-reconfiguration`, b790e8d).
Status	All four features tested in the sim by the user: the failures and the A380X reconfiguration OK, the EWD and SD on the ND display unit included (load, CAPT, F/O, both sides); the A32NX reconfiguration OK on all steps. The A380X mouse fix for DUs holding hidden displays: OANS clicks on the F/O ND OK on a combined test build, not sim-tested on this core branch. Hosted display gate and CAPT PFD FMS side (added): not sim-tested, unit tests only.
Pictures	No pictures: a failed display unit is blank, the alerts are the stock FBW E/WD and SD displays.

1. A32NX ND and ECAM display unit failures
2. A32NX display unit reconfiguration
3. A380X CDS display unit failures
4. A380X display unit reconfiguration
5. A380X hosted EWD and PFD: hidden when not shown
6. A380X CAPT PFD reads the CAPT side FMS data

1. A32NX ND and ECAM display unit failures

> **Aircraft:** A32NX | **Where:** flyPad > Failures > ATA 31; CPT / F/O ND, upper and lower ECAM | **Status:** Tested in the sim by the user: OK.

What it is for

The flyPad could fail the two PFDs only. Four more display units (DUs) can now be failed: the CAPT and F/O NDs and the upper and lower ECAM DUs. A failed DU goes blank, the way a failed PFD already did. The A320 FCOM (DSC-31-05-60, FAILURE OF A DU) lists a blank screen with an amber F, a distorted display or a blank screen with INVALID DISPLAY UNIT; the blank (OFF) screen is used here.

- The weather radar and terrain picture of the ND (the native gauge over it) goes blank with a failed ND.
- It is also no longer drawn on an ND that is switched off (brightness knob OFF) or still in its self-test.
- The lower ECAM failure blanks the whole SD (both of its display layers). Before, the SD display unit ignored a failure change.
- When the failure is cleared, the DU runs its self-test, then shows its picture again.

The flyPad failures (ATA 31 Indicating / Recording System)

flyPad failure name	Display unit that goes blank
Captain PFD display	CAPT PFD (existing failure)
F/O PFD display	F/O PFD (existing failure)
Captain ND display	CAPT ND, with its weather / terrain picture

F/O ND display	F/O ND, with its weather / terrain picture
Upper ECAM display	Upper ECAM DU (E/WD)
Lower ECAM display	Lower ECAM DU (SD)

Limitations and why

Limitation	Why	What to do
No ECAM alert for a failed DU.	The branch only blanks the DU.	-
A lost DU's picture moves only as the FCOM says.	See "A32NX display unit reconfiguration" below: the PFD goes to the ND DU and the E/WD to the lower ECAM DU by themselves; the ND and the SD need the PFD/ND XFR pb, the ECAM/ND XFR selector or a held system page key (FCOM DSC-31-05-60).	Use those controls.
Always a blank screen.	The amber F, the distorted picture and INVALID DISPLAY UNIT of the FCOM are not drawn.	-

How to use

1. Open **flyPad > Failures**, chapter **ATA 31 Indicating / Recording System**.
2. Activate one of the display failures (table above): that DU goes blank.
3. Deactivate it to get the DU back (after its self-test).

How to test

#	Do this	Expected result
1	A32NX, any phase, all displays on. CPT ND in ARC with the weather radar on (weather around) or TERR ON ND. flyPad > Failures > ATA 31: activate **Captain ND display**.	The CPT ND goes blank, with no weather or terrain left on it. The F/O ND is not affected.
2	Deactivate **Captain ND display**.	The CPT ND shows its self-test, then the ND picture with the weather / terrain.
3	Activate **F/O ND display**, then deactivate it.	Same on the F/O ND.
4	Activate **Upper ECAM display**.	The upper ECAM DU goes blank; the E/WD moves to the lower ECAM DU (automatic reconfiguration, next section).
5	Deactivate it; activate **Lower ECAM display**.	The E/WD comes back on the upper DU after its self-test; the lower ECAM DU (SD) goes blank, whatever SD page was shown.
6	Deactivate it.	The SD comes back after its self-test.
7	No failure. With the weather radar picture on the CPT ND, turn the CPT ND brightness knob fully OFF, then back on.	The ND and its weather picture go dark together (no weather left on the dark ND) and come back together.

2. A32NX display unit reconfiguration

> **Aircraft:** A32NX | **Where:** PFD/ND XFR pbs; PFD and ND brightness knobs; SWITCHING panel ECAM/ND XFR; ECAM control panel page keys | **Status:** Tested in the sim by the user on all steps: OK. Unit tests: DisplayReconfiguration.spec.ts (shared), DmcDisplayReconfiguration.spec.ts (systems host), displayUnit.spec.tsx (common DU and legacy SD).

What it is for

The A320 moves the picture of a lost display unit (DU) to another one (A320 FCOM DSC-31-05-60 RECONFIGURING DUS). Before, a failed or switched-off A32NX DU simply lost its picture. Now the PFD, the ND, the E/WD and the SD can be shown on another DU, automatically or with the cockpit controls, as the FCOM describes.

> **NOTE**
> A320 FCOM DSC-31-05-60 : if the upper ECAM display fails or is switched off, "the engine/warning page automatically replaces the system/status page on the lower ECAM DU"; "If a PFDU fails, the system automatically transfers the PFD image to the NDU."

> **NOTE**
> A320 FCOM DSC-31-50 : turning the PFD brightness knob fully counterclockwise switches off the PFD, its image goes to the NDU, and the pilot can recover the ND with the PFD/ND XFR pushbutton. DSC-31-30 : the ECAM/ND XFR selector moves the system/status display to the CAPT or F/O ND with "ECAM ON ND" on the lower ECAM display; with both ECAM DUs failed it moves the E/WD, without that message.

What you should see

Situation	What you should see
PFD DU lost (flyPad failure, PFD brightness knob OFF, or no power)	The PFD DU is blank and the PFD is shown on the ND DU of the same side (the ND is lost until the PFD/ND XFR pb is pressed).
PFD/ND XFR pb pressed	The PFD and the ND of that side swap DUs; press again to swap back. After an automatic transfer, the pb brings the ND back on the ND DU (FCOM DSC-31-50).
ND DU lost	Nothing automatic; the PFD/ND XFR pb puts the ND on the PFD DU (FCOM).
Upper ECAM DU lost	The E/WD replaces the SD on the lower ECAM DU.
Lower ECAM DU lost	Nothing automatic; the SD is lost until ECAM/ND XFR or a held page key is used.
ECAM/ND XFR on CAPT (F/O)	The SD is shown on the CAPT (F/O) ND DU; the lower ECAM DU shows **ECAM ON ND**. The SWITCHG PNL memo comes on only if a PFD/ND XFR pb is pressed as well (FCOM DSC-31-30 MEMO DISPLAY, the EIS DMC selector alone still shows it).
Both ECAM DUs lost, ECAM/ND XFR on CAPT (F/O)	The E/WD is shown on that ND DU, with no ECAM ON ND message.
A system page key held (0.5 s) with one ECAM DU lost	The SD (the page of that key) is shown on the remaining ECAM DU instead of the E/WD, for at most 3 minutes; releasing the key brings the E/WD back. With both ECAM DUs lost, it shows the SD on the ND chosen with ECAM/ND XFR.
Lost DU recovered	Each picture goes back to its own DU; the recovered DU runs its self-test first.

How it is done

No extra copies of the instruments. MSFS draws a gauge on the model material of the same name, so every DU has one glass per picture it can show (17 overlay glasses carrying the material of that picture, an ECAM ON ND glass and a blank glass). The DMC logic of the systems host writes the picture of each DU, and the cockpit behaviour shows the matching glass. The brightness stays the DU's own knob. A new small gauge (`DuMessage`) draws the ECAM ON ND message.

Design choices

- The DU that receives a picture does not run a self-test: the picture is already live (the FCOM gives none).

- A system page key must be held 0.5 s before the SD appears: a click that only selects or deselects a page does not flash the SD, and the SD has its new page drawn before it is shown.
- ECAM ON ND is drawn in the style of the SELF TEST IN PROGRESS message: the FCOM gives no colour.
- A lost DU shows dark glass (a blank screen, which the FCOM allows for a failed DU).
- An unpowered DU counts as unavailable, like a failed one or one switched off with its knob, so its picture moves.
- A change of the PFD DU (lost or back) restarts from the automatic configuration: the PFD always goes to an available DU first.

Limitations and why

Limitation	Why	What to do
No DMC failures, no INVALID DATA, the EIS DMC selector only feeds the memo.	The three DMCs are not modelled (FCOM DSC-31-05-50).	-
No STS re-press within 2 s for the next STATUS page and no advisory key-light pulse in the single-display mode.	Not part of this change; there is no ECAM advisory logic yet.	-
An ECAM picture on an ND DU is replaced by the PFD when that side's PFD DU is lost or its PFD/ND XFR pb is pressed.	Design choice: the PFD keeps the priority on the ND DU; the picture moved to the PFD DU is always the ND.	Use the other ND with ECAM/ND XFR.
Needs the cockpit model companion and an MSFS restart.	The glasses are in the A32NX interior model (aircraft-large-files fork branch `feat/a32nx-du-reconfiguration`, b790e8d, stacked on `feat/a32nx-printer-door` 99237fb and the F/O flyPad screen d3060ea); panel.cfg (new `DuMessage` gauge, `[VCockpit21]` on this branch) and the behaviour XML change.	Install the model companion and restart the sim. With `core/efb-redesign` also merged, renumber one of the two `[VCockpit21]` sections.

How to use

1. Nothing to do for the automatic transfers: a lost PFD DU or upper ECAM DU moves its picture by itself.
2. Press the **PFD/ND XFR** pushbutton of your side to swap your PFD and ND (or to get the ND back after an automatic PFD transfer).
3. To see the SD with an ECAM DU lost: turn the **ECAM/ND XFR** selector on the SWITCHING panel to CAPT or F/O, or hold the system page key on the ECAM control panel.
4. Turn the ECAM/ND XFR selector back to NORM to get the ND back.

How to test

#	Do this	Expected result
1	A32NX powered, all displays on. flyPad > Failures > ATA 31: activate **Captain PFD display**.	The CAPT PFD DU goes blank; the PFD appears on the CAPT ND DU at once (no self-test there).
2	Press the CAPT **PFD/ND XFR** pushbutton.	The ND comes back on the CAPT ND DU; the PFD DU stays blank.
3	Deactivate the failure.	The CAPT PFD DU runs its self-test and shows the PFD; the ND DU shows the ND.
4	Press the CAPT PFD/ND XFR pb, then again.	The PFD and ND swap DUs, then swap back.

5	Turn the F/O PFD brightness knob fully OFF, then back on.	The PFD appears on the F/O ND DU, then goes back to the PFD DU.
6	Activate **Upper ECAM display**.	The upper ECAM DU goes blank; the E/WD replaces the SD on the lower ECAM DU.
7	Hold a system page key (e.g. **HYD**) on the ECAM control panel, then release it.	After about half a second the HYD page is shown on the lower ECAM DU; on release the E/WD comes back. A short click does not flash the SD.
8	Turn the **ECAM/ND XFR** selector to CAPT.	The SD is shown on the CAPT ND DU; the lower ECAM DU keeps the E/WD (the upper DU is still lost).
9	Deactivate the failure, keep ECAM/ND XFR on CAPT.	The E/WD is back on the upper DU; the lower ECAM DU shows **ECAM ON ND**; no SWITCHG PNL memo.
10	Press the CAPT PFD/ND XFR pb.	The CAPT PFD and ND swap DUs (the PFD replaces the SD on the CAPT ND DU, design choice) and the SWITCHG PNL memo appears in green. Press it again: the SD is back on the CAPT ND DU and the memo goes.
11	Activate **Upper ECAM display** and **Lower ECAM display**, ECAM/ND XFR on F/O.	The E/WD is shown on the F/O ND DU, without ECAM ON ND. Holding a page key shows the SD there instead.
12	Deactivate all failures, ECAM/ND XFR to NORM.	Every DU shows its own picture again.

3. A380X CDS display unit failures

> **Aircraft:** A380X | **Where:** flyPad > Failures > ATA 31; the 8 CDS DUs; E/WD; SD STATUS | **Status:** Tested in the sim by the user: OK.

What it is for

The eight display units of the A380 CDS (A380 FCOM DSC-31-15-10: CAPT PFD, CAPT ND, CAPT MFD, E/WD, SD, F/O MFD, F/O ND, F/O PFD) can be failed from the flyPad. The failed DU goes blank (on an ND, the weather radar and terrain picture too, the VD included) and the FWS raises the **CDS ... DU FAULT** alert with its FCOM procedure and its INOP SYS line (FCOM PRO-ABN-ECAM-10-31).

The flyPad failures (ATA 31 Indicating / Recording System)

flyPad failure name	ECAM alert (amber)	STATUS INOP SYS
CAPT PFD display unit	CDS CAPT PFD DU FAULT	CAPT PFD DU
CAPT ND display unit	CDS CAPT ND DU FAULT	CAPT ND DU
CAPT MFD display unit	CDS CAPT MFD DU FAULT	CAPT MFD DU
EWD display unit	CDS EWD DU FAULT	EWD DU
SD display unit	CDS SD DU FAULT	(none: the FCOM lists no SD line)
F/O MFD display unit	CDS F/O MFD DU FAULT	F/O MFD DU
F/O ND display unit	CDS F/O ND DU FAULT	F/O ND DU
F/O PFD display unit	CDS F/O PFD DU FAULT	F/O PFD DU

What you should see

Situation	What you should see

Failure activated	The DU goes blank. Single chime, MASTER CAUT, and the amber alert on the E/WD.
ECAM procedure (each DU)	*xx* DU ..... OFF 5 S THEN ON; IF NOT SUCCESSFUL: *xx* DU ..... OFF; DU RECONF P/B AVAIL (green). The SD DU FAULT adds: IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX.
SD STATUS page	INOP SYS (all phases): the DU line of the table above.
Takeoff and landing	The alert is inhibited in the FWS flight phases 3 to 7, 9 and 10; the DU is blank anyway.
DU without power	No DU FAULT alert (design choice): the lost busbar has its own ELEC alert.
Failure cleared	The DU runs its self-test and shows its picture again; the alert and the INOP SYS line go.

Limitations and why

Limitation	Why	What to do
The reconfiguration has limits.	See "A380X display unit reconfiguration" below: the PFD and the EWD move to another DU by themselves, the MFD never does, and the SD only on an ND DU with the DU RECONF pb, without the ATC mailbox.	With the SD DU failed, press the F/O DU RECONF pb (it acts on the F/O ND DU) until the SD pages (STATUS included) show there.
The OFF 5 S THEN ON item does not bring the DU back.	The flyPad failure stays until it is cleared, which is the procedure's NOT SUCCESSFUL branch.	Clear the failure in the flyPad.
The second FCOM trigger (a DU that declared two other DUs faulty in turn) is not modelled.	It needs the DU monitoring loops, which are not simulated.	-

How to use

1. Open **flyPad > Failures**, chapter **ATA 31 Indicating / Recording System**.
2. Activate one of the eight **... display unit** failures.
3. Follow the ECAM procedure, then check the SD STATUS page.
4. Deactivate the failure to get the DU back.

How to test

#	Do this	Expected result
1	A380X on the ground (before the takeoff power) or in cruise, all DUs on. CPT ND in ARC with the WX overlay and weather around. flyPad > Failures > ATA 31: activate **CAPT ND display unit**.	The CAPT ND and its VD go blank, with no weather or terrain left. Single chime, MASTER CAUT. E/WD: **CDS CAPT ND DU FAULT** with CAPT ND DU ..... OFF 5 S THEN ON, IF NOT SUCCESSFUL, CAPT ND DU ..... OFF, DU RECONF P/B AVAIL in green.
2	Open the SD **STATUS** page.	INOP SYS: **CAPT ND DU**.
3	Deactivate the failure.	The CAPT ND runs its self-test, then shows the ND, the VD and the weather again; the alert and the INOP SYS line are gone.
4	Repeat with **CAPT PFD display unit**, **CAPT MFD display unit**, **F/O PFD display unit**, **F/O ND display unit** and **F/O MFD display unit**.	Each time the named DU goes blank and the E/WD shows the matching CDS ... DU FAULT, with its INOP SYS line on the STATUS.

5	Activate **SD display unit**.	The SD goes blank; E/WD: **CDS SD DU FAULT** with the extra green line IF SD DU OFF : F/O DU RECONF P/B FOR ATC MAILBOX (the FCOM gives this alert no INOP SYS line).
6	Deactivate it; activate **EWD display unit**.	The EWD DU goes blank and the E/WD moves to the SD DU (automatic reconfiguration), where CDS EWD DU FAULT can be read; single chime and MASTER CAUT; INOP SYS **EWD DU** on the STATUS.
7	Activate any DU failure during the takeoff roll (above takeoff power).	The DU goes blank but no alert until the end of the inhibition.

4. A380X display unit reconfiguration

> **Aircraft:** A380X | **Where:** CAPT and F/O PFD/ND and DU RECONF pushbuttons; the 8 CDS DUs | **Status:** Tested in the sim by the user on all steps: OK. Mouse fix: OANS clicks on the F/O ND OK and the ATC mailbox on the F/O ND (needs `core/a380x-mailbox-du-reconf`) OK, on a combined test build; not sim-tested on this core branch. Unit tests: CdsReconfiguration.spec.ts (shared and systems host), HostedDisplay.spec.ts, hostedInstrument.spec.ts; native ndwxr test cds_display_test.

What it is for

The A380 CDS moves a lost display to another display unit (A380 FCOM DSC-31-15-20 CDS abnormal operations, DSC-31-20-50 PFD/ND pb). Before, a failed or switched-off DU simply lost its picture. Now:

- **Automatic:** when a PFD DU fails, loses its power or is switched off, the PFD is shown on the ND DU of the same side (the ND is lost). When the EWD DU is lost, the E/WD is shown on the SD DU.
- **PFD/ND pushbutton:** exchanges the on-side PFD and ND. With the exchange selected, the PFD is on the ND DU and comes back to the PFD DU if the ND DU fails (the PFD keeps the priority).
- **DU RECONF pushbutton:** active only with a failed on-side DU; it acts on the DU the FCOM gives (CAPT: the MFD DU, then the ND DU if the MFD DU is failed, then the PFD DU if the ND and MFD DUs are failed; F/O: the ND DU, the MFD DU if the PFD or ND DU is failed, then the PFD DU) and steps through the displays of the FCOM order that this DU can draw, then back to its normal display.
- **EWD and SD on the ND DU:** with the CAPT (F/O) PFD and MFD DUs failed, the pb acts on the ND DU, which shows the PFD automatically; it now cycles **PFD > EWD > SD > PFD** (the MFD of the FCOM sequence cannot be drawn there). With only the MFD DU (CAPT) or the SD DU (F/O) failed, the ND DU cycles ND > EWD > SD > ND the same way. The EWD (SD) is never shown on both sides at once (A380 FCOM DSC-31-15-20), and a PFD that moves to the ND DU later (its PFD DU fails afterwards) wins over a selection there.
- The weather radar and terrain picture (ndwxr) is hidden while its ND DU shows another display (the PFD, the EWD or the SD).

What you should see

Situation	What you should see
CAPT PFD display unit failed (or switched off)	The CAPT PFD DU is blank and the PFD appears on the CAPT ND DU, without weather or terrain on it.
EWD display unit failed	The E/WD appears on the SD DU.
PFD/ND pb pressed (CAPT or F/O)	The PFD and the ND of that side swap DUs; press again to swap back.
CAPT ND and MFD DUs failed, DU RECONF pressed	The CAPT PFD DU steps from the PFD to the ND (with no weather or terrain image), then back to the PFD.

CAPT PFD and MFD DUs failed, DU RECONF pressed	The CAPT ND DU, which shows the PFD, steps to the E/WD, then to the SD, then back to the PFD. While the E/WD (SD) is shown there, the F/O pb skips it.
SD DU failed, F/O DU RECONF pressed	The F/O ND DU steps from the ND to the E/WD, then to the SD pages (without the ATC mailbox), then back to the ND.
Failure cleared	Each DU goes back to its normal display.

The mouse on a DU with hidden displays

With the reconfiguration, every DU also holds the gauges of the displays it can show (the ND DUs hold a PFD, an E/WD and an SD as well). A hidden gauge stayed on top and took the mouse, so the OANS clicks on the F/O ND did not work (and the ATC mailbox on an ND DU of `core/a380x-mailbox-du-reconf`). Now the hosted E/WD and PFD never take the mouse, the hosted SD takes it only while it is shown, and the ND keeps it (OANS). The build adds one CSS rule per hosted gauge, and the display gate of each DU marks the hidden ones, found by their panel.cfg URL.

Limitations and why

Limitation	Why	What to do
The MFD is never drawn on another DU, and nothing is drawn on the MFD DUs; the DU RECONF pb skips them, so it does nothing when it acts on an MFD DU.	Every display is a gauge of its own on its DU texture: the MFD is one gauge over both MFD screens (one texture) that also runs the FMS. Design choice: the PFD and the ND on both DUs of a side, the E/WD and the SD also on the ND DUs, and the E/WD on the SD DU.	Use the other side's MFD.
The SD on an ND DU has no ATC mailbox.	The mailbox stays with the SD DU's own SD (design choice of the hosted SD). `core/a380x-mailbox-du-reconf` brings it to the ND DU.	With the SD DU failed, the ATC mailbox is lost on this branch alone.
A selected E/WD or SD replaces the PFD on that side.	With the PFD and MFD DUs of a side failed, the ND DU is the only DU left for the PFD (FCOM: the pb acts on the ND DU).	Press DU RECONF again to get the PFD back.
The ND on the PFD DU has no weather radar or terrain image.	The native ndwxr gauges stay on the ND DUs.	Use the other side's ND.
The KCCU keys do not reconfigure the DUs.	Not modelled.	Use the DU RECONF pb.
A full MSFS restart is needed after an update.	panel.cfg (new hosted gauges, the E/WD and the SD on the ND DUs included) and the cockpit behaviour XML change.	Restart the sim after installing.

How to use

1. Nothing to do for the automatic reconfiguration: a lost PFD or EWD DU moves its display by itself.
2. Press the **PFD/ND** pushbutton of your side to exchange your PFD and ND.
3. With a failed on-side DU, press **DU RECONF** to step the display of the DU it acts on.
4. With your PFD and MFD DUs failed, press **DU RECONF** to bring the E/WD, then the SD, onto your ND DU; once more for the PFD.

How to test

#	Do this	Expected result

1	A380X powered, all DUs on. flyPad > Failures > ATA 31: activate **CAPT PFD display unit**.	The CAPT PFD DU goes blank; the PFD appears on the CAPT ND DU (no weather / terrain image on it); CDS CAPT PFD DU FAULT.
2	Deactivate it. Press the CAPT **PFD/ND** pushbutton.	The CAPT PFD and ND swap DUs. Press again: back to normal.
3	Turn the F/O PFD DU brightness knob OFF.	The PFD appears on the F/O ND DU. Turn it back on: normal again.
4	Activate **EWD display unit**.	The E/WD appears on the SD DU. Deactivate it: the SD comes back.
5	Activate **CAPT ND display unit** and **CAPT MFD display unit**, then press the CAPT **DU RECONF** pushbutton twice.	The CAPT PFD DU shows the ND, then the PFD again.
6	Deactivate them. Activate **CAPT PFD display unit** and **CAPT MFD display unit**.	The PFD appears on the CAPT ND DU.
7	Press the CAPT **DU RECONF** pushbutton three times.	The CAPT ND DU shows the E/WD, then the SD (the current SD page), then the PFD again. No weather or terrain image on it meanwhile.
8	Press the CAPT DU RECONF pb once (E/WD on the CAPT ND DU). Then activate **F/O PFD display unit** and **F/O MFD display unit** and press the F/O **DU RECONF** pb.	The F/O ND DU skips the E/WD (shown on the CAPT side) and shows the SD.
9	Deactivate all failures. Activate **SD display unit** and press the F/O **DU RECONF** pb twice.	The F/O ND DU shows the E/WD, then the SD pages (no ATC mailbox); once more: the ND again.
10	Deactivate all failures.	Every DU shows its normal display.
11	No failure, on the ground at an airport: show the OANS airport map on the F/O ND (F/O EFIS control panel, range ZOOM). Drag the map with the mouse and click a runway or taxiway label.	The map pans and the click is taken on the F/O ND (before the mouse fix nothing reacted there).

5. A380X hosted EWD and PFD: hidden when not shown

> **Aircraft:** A380X | **Where:** SD DU and both ND DUs (where the EWD and PFD can be shown after a DU RECONF) | **Status:** Not sim-tested (added; unit tests: EWD/instrument.spec.ts 3 and PFD/instrument.spec.ts 2, loading the real instruments in a sim-like gauge panel; 3 fail before the change).

What it is for

The EWD and the PFD also run as hidden copies on the display units that can take them over (EWD: SD DU and both ND DUs; PFD: both ND DUs). Each copy has a gate that hides it and stops it while its DU shows its own display. The gate was never made: the instruments looked up their display unit before the sim had connected the gauge, found the wrong one and made no gate. The hidden copies then ran and rendered all the time and were never hidden. The gate is now made once the gauge is connected.

Situation	What you should see
Normal configuration	The hidden EWD and PFD copies are hidden and stopped on the SD and ND DUs.
EWD DU failed (or DU RECONF), EWD on the SD DU or an ND DU	The EWD copy shows there, as before.
PFD shown on the ND DU	The PFD copy shows there, as before.

Limitations and why

Limitation	Why	What to do
No visible change in the normal configuration.	The fix stops hidden copies; the effect is less work for the sim and no hidden copy in front of the DU.	-

How to use

1. Nothing to set: use the DU RECONF and PFD/ND pushbuttons as in the reconfiguration feature above.

How to test

#	Do this	Expected result
1	A380X powered, no failure. Use the SD mailbox buttons and click on the ND / OANS.	The clicks are taken (no hidden copy in front).
2	flyPad > Failures > ATA 31: **EWD display unit**.	The EWD shows on the SD DU.
3	Remove the failure; press the CAPT PFD/ND pb (PFD on the ND DU), then again.	The PFD shows on the ND DU, then the ND comes back.
4	Press a DU RECONF pb through its cycle.	Each display shows on the ND DU and hides again.

What PR testers should check

- SD mailbox and ND / OANS clicks.
- The EWD on the SD DU after an EWD DU failure; the PFD on the ND DU; the DU RECONF cycle.
- Not sim-tested yet.

6. A380X CAPT PFD reads the CAPT side FMS data

> **Aircraft:** A380X | **Where:** CAPT PFD (also when drawn on the CAPT ND DU) | **Status:** Not sim-tested (added; unit tests: PFD/instrument.spec.ts, 4 side cases, the CAPT case failing before).

What it is for

The captain PFD picked its FMS side before the sim had connected the gauge, and the side number it used was never 1: the CAPT PFD read the F/O side FMS data (the FM message flags, the TO waypoint, the MRP and the RNP). The side is now set when the gauge is connected, from the PFD's own display index (1 = CAPT, 2 = F/O, also when the PFD is drawn on an ND DU). The same bug is in FlyByWire upstream. The A32NX PFD is not affected.

Limitations and why

Limitation	Why	What to do
No visible change while both FMS sides agree.	The two sides normally show the same data.	Make them differ to see it (test below).

How to use

1. Nothing to set.

How to test

#	Do this	Expected result

1	A380X in flight on a flight plan. Set a different required RNP or FM message on each side (for example an RNP entered on the CAPT side only), then compare both PFDs.	The CAPT PFD shows the CAPT side data, the F/O PFD the F/O side data (before: both the F/O side).
2	CAPT PFD/ND pb: PFD on the CAPT ND DU.	Still the CAPT side data.

What PR testers should check

- The CAPT PFD FMA messages, RNP and TO waypoint match the CAPT FMS side.
- Not sim-tested yet.